

# TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

## PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 06

**75 minutes**

### INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

**Calculators and dictionaries are NOT permitted.**

**There is no formulae booklet for this test.**

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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- 1 For what value of  $c$  does the line  $y = 2x + c$  touch the parabola  $y = x^2 - 4x + 7$ ?
- A 7
  - B 2
  - C -2
  - D -1
  - E 0
  - F 5

- 2 A curve has  $\frac{dy}{dx} = 6x^2 - 4x + 1$  and passes through  $(1, 3)$ . Find its equation.
- A**  $y = 12x - 4$
- B**  $y = 2x^3 - 4x^2 + x + 4$
- C**  $y = 2x^3 - 2x^2 + x + 2$
- D**  $y = 2x^3 - 2x^2 + x + 3$
- E**  $y = 2x^3 - 2x^2 + x$
- F**  $y = 2x^3 - 2x^2 + 2$

**3** Which of the following is an identity (true for all values of  $x$ )?

**A**  $(x + 3)^2 - (x - 3)^2 = 12x$

**B**  $x^2 + 6x + 9 = 0$

**C**  $(x + 3)^2 = x^2 + 9$

**D**  $(x - 3)^2 = x^2 - 6x - 9$

**E**  $(x + 3)(x - 3) = x^2 + 9$

- 4 How many real solutions does the equation  $2^{x^2} = 16 \times 2^{-3x}$  have?
- A 3
  - B 2
  - C 4
  - D 1
  - E 0
  - F infinitely many

5 Find  $\frac{dy}{dx}$  when  $y = \frac{(3x-2)^2}{x^2}$ .

**A**  $\frac{12}{x^2} + \frac{8}{12x^3}$

**B**  $\frac{6}{x} - \frac{12x^2}{8}$

**C**  $-\frac{12x^2}{6} + \frac{8}{12x^3}$

**D**  $\frac{12x^2}{6(3x-2)} - \frac{8x^3}{12}$

**E**  $\frac{x^2}{9} - \frac{12}{x}$

**F**  $9 - \frac{x^2}{x}$

**6** Find the equation of the line through  $(2, -3)$  that is perpendicular to  $3x - 2y + 5 = 0$ .

**A**  $2x - 3y - 13 = 0$

**B**  $3x + 2y = 0$

**C**  $2x + 3y + 5 = 0$

**D**  $2x + 3y - 13 = 0$

**E**  $2x + 3y - 5 = 0$

**F**  $3x - 2y - 12 = 0$

7 Find the radius of the circle  $x^2 + y^2 - 6x + 4y - 12 = 0$ .

**A** 4

**B** 5

**C** 1

**D**  $\sqrt{12}$

**E**  $\sqrt{13}$

**F** 25

**8** Find the sum of the y-coordinates of the stationary points of the curve  $y = 2x^3 - 9x^2 + 12x - 3$ .

**A** 2

**B** 0

**C** 5

**D** 4

**E** 3

**F** 1

**9** A sequence is defined by  $x_{n+1} = 3x_n - 2$  with  $x_1 = 2$ . Find  $x_5$ .

**A** 244

**B** 80

**C** 28

**D** 82

**E** 162

**F** 54

**10** Two fair dice are thrown. Find the probability that the product of the two scores is odd.

- A**  $\frac{2}{3}$
- B**  $\frac{4}{3}$
- C**  $\frac{1}{4}$
- D**  $\frac{1}{6}$
- E**  $\frac{2}{3}$
- F**  $\frac{1}{3}$

**11** Find the coefficient of  $x^5$  in the expansion of  $(1+x)^3(1+x)^4$ .

**A** 35

**B** 21

**C** 12

**D** 1

**E** 28

**F** 7

**12**  $A$ ,  $B$  and  $C$  lie on a circle with centre  $O$ . Angle  $AOB = 100^\circ$  and  $C$  lies on the major arc  $AB$ . Find angle  $ACB$ .

**A**  $200^\circ$

**B**  $100^\circ$

**C**  $40^\circ$

**D**  $50^\circ$

**E**  $130^\circ$

**F**  $80^\circ$

- 13** The graph of  $y = x^2$  is translated by 3 units in the positive  $x$ -direction and then stretched parallel to the  $y$ -axis with scale factor 2. What is the equation of the resulting graph?

**A**  $y = 2(x - 3)^2$

**B**  $y = (x - 3)^2 + 2$

**C**  $y = 2(x + 3)^2$

**D**  $y = 2x^2 - 3$

**E**  $y = (2x - 3)^2$

**F**  $y = \frac{1}{2}(x - 3)^2$

**14** Solve  $2^{2x} - 3 \cdot 2^{x+2} + 32 = 0$ . Find the sum of the solutions.

**A** 6

**B** 12

**C** 4

**D** 3

**E** 8

**F** 5

- 15** A regular hexagon and a regular octagon share a common side. What is the size of the angle between their two other sides that meet at one end of the common side (the angle outside both polygons)?
- A**  $120^\circ$
  - B**  $95^\circ$
  - C**  $105^\circ$
  - D**  $135^\circ$
  - E**  $115^\circ$
  - F**  $75^\circ$

- 16** The curve  $y = x(x - 1)(x - 2)$  and the  $x$ -axis enclose two regions. What is the ratio of the area of the region for  $0 \leq x \leq 1$  to the area for  $1 \leq x \leq 2$ ?
- A**    1 : 2
- B**    2 : 1
- C**    1 : 3
- D**    1 : 4
- E**    1 : 1
- F**    3 : 1

- 17 The expression  $x^4 + 4$  can be factorised into two quadratics with integer coefficients. What is the sum of the coefficients of  $x$  in these two quadratics?
- A 1
  - B 0
  - C 2
  - D 4
  - E  $-4$
  - F It cannot be factorised.

- 18  $\vec{AB} = 3\mathbf{p} + 2\mathbf{q}$  and  $\vec{BC} = k\mathbf{p} - 4\mathbf{q}$ , where  $\mathbf{p}$  and  $\mathbf{q}$  are not parallel. Given that  $A$ ,  $B$  and  $C$  are collinear, find  $k$ .

A 2

B 6

C -3

D  $\frac{3}{2}$

E -6

F -4

- 19** In triangle  $ABC$ ,  $AB = 6$ ,  $BC = 4$  and angle  $BAC = 30^\circ$ . How many possible triangles are there?
- A** 1
  - B** 3
  - C** 2
  - D** 0
  - E** infinitely many

**20** Which one of the following is the largest?

**A**  $2\sqrt{4.5}$

**B**  $\sqrt{2} + \sqrt{7}$

**C**  $\sqrt{3} + \sqrt{6}$

**D**  $\sqrt{4} + \sqrt{5}$

**E**  $\sqrt{1} + \sqrt{8}$

**END OF TEST**